

What the raw data contains

A look at the three hospital extracts we were given (lab results, medications and clinical notes) before any cleaning: how big they are, who they cover, how far back they go, and where the gaps are.

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01 The three files

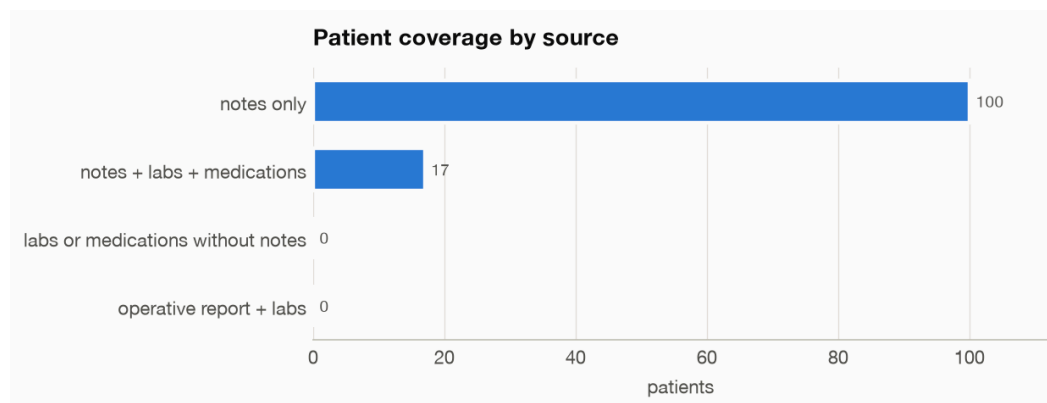
Each file is a single spreadsheet exported from the hospital record system. Patients are pseudonymised and every date is shifted and cut down to the year, so the finest time step we have is one year.

File	Rows	Columns	Patients	Rows per patient (median)	Years	Repeated rows
Lab results	43,550	16	17	1,775	2004 to 2027	56%
Medications	5,807	23	17	222	2001 to 2027	63%
Clinical notes	215	11	117	2	2006 to 2026	none

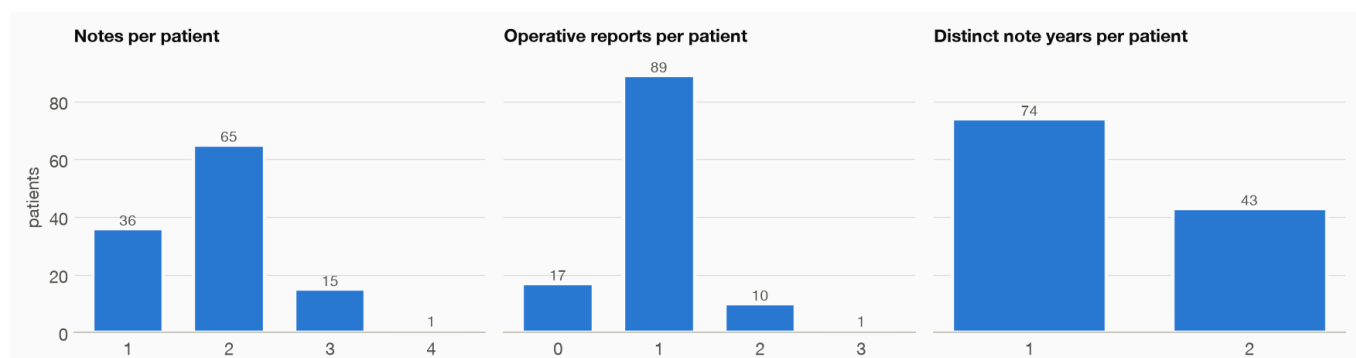
Some rows are dated 2027, a year that has not happened yet. The date shift used for anonymisation pushed them there.

02 Which patients each file covers

The notes cover all 117 patients. The lab and medication files cover the same 17 patients, and those 17 have a single note each. None of them has an operative report.



So the data splits into two groups that never overlap. For 100 patients we know the operation but have no lab or medication history. For the other 17 we have the history but no operative report.



Most patients have one or two notes. 89 have exactly one operative report. No patient has notes from more than two different years.

03 Records over time

The three files differ in size by a factor of 200, so each has its own panel. For labs and medications the grey part of each bar is repeated rows and the blue part is what is left after removing them.



Lab and medication rows peak in 2010 to 2013 and again in 2023. Operative reports run from 2006 to 2022. After 2022 only follow-up notes remain.

The spikes in the lab and medication panels line up with hospital admissions. Most of those rows come from hospital stays around surgery, not from years of follow-up.

04 Repeated rows

A row counts as repeated when every column matches an earlier row exactly. More than half of each structured file is repetition, and a single lab row appears 826 times.

19,002

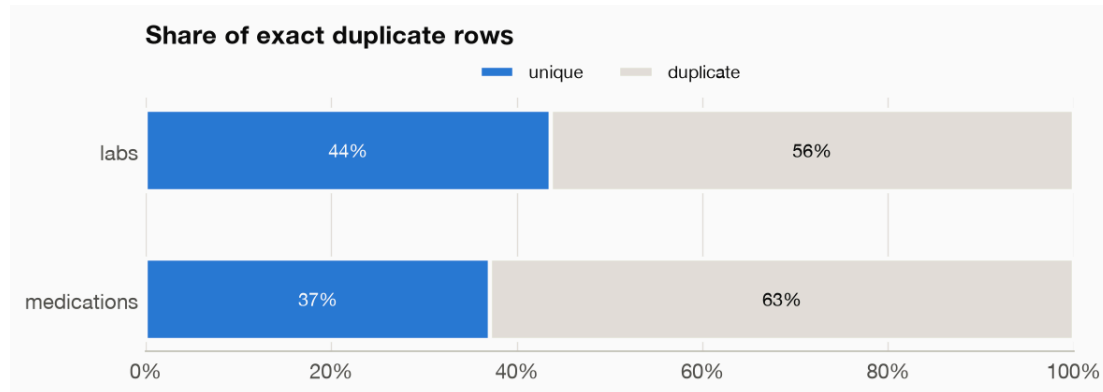
distinct lab rows (of 43,550)

2,157

distinct medication rows (of 5,807)

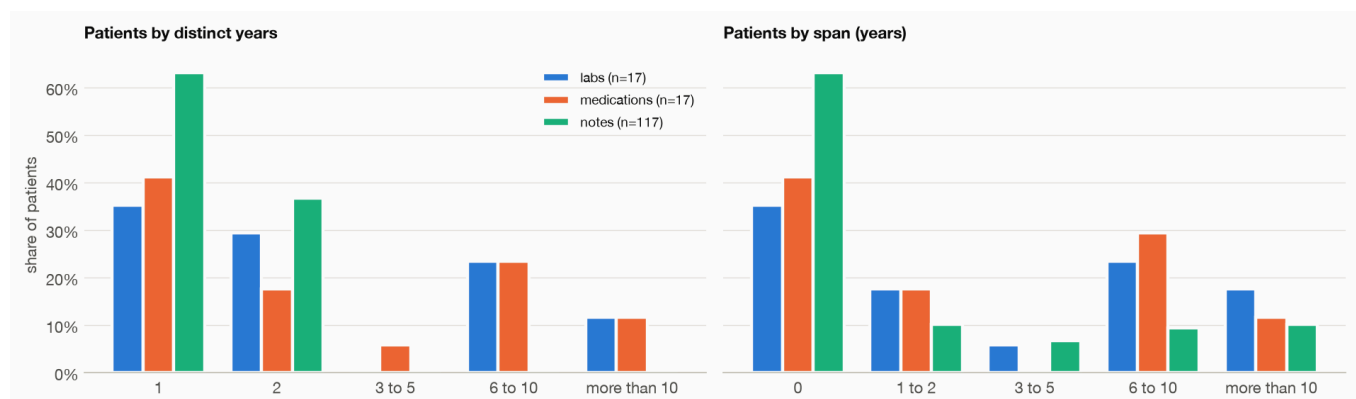
826

copies of the most repeated lab row



05 How long patients are followed

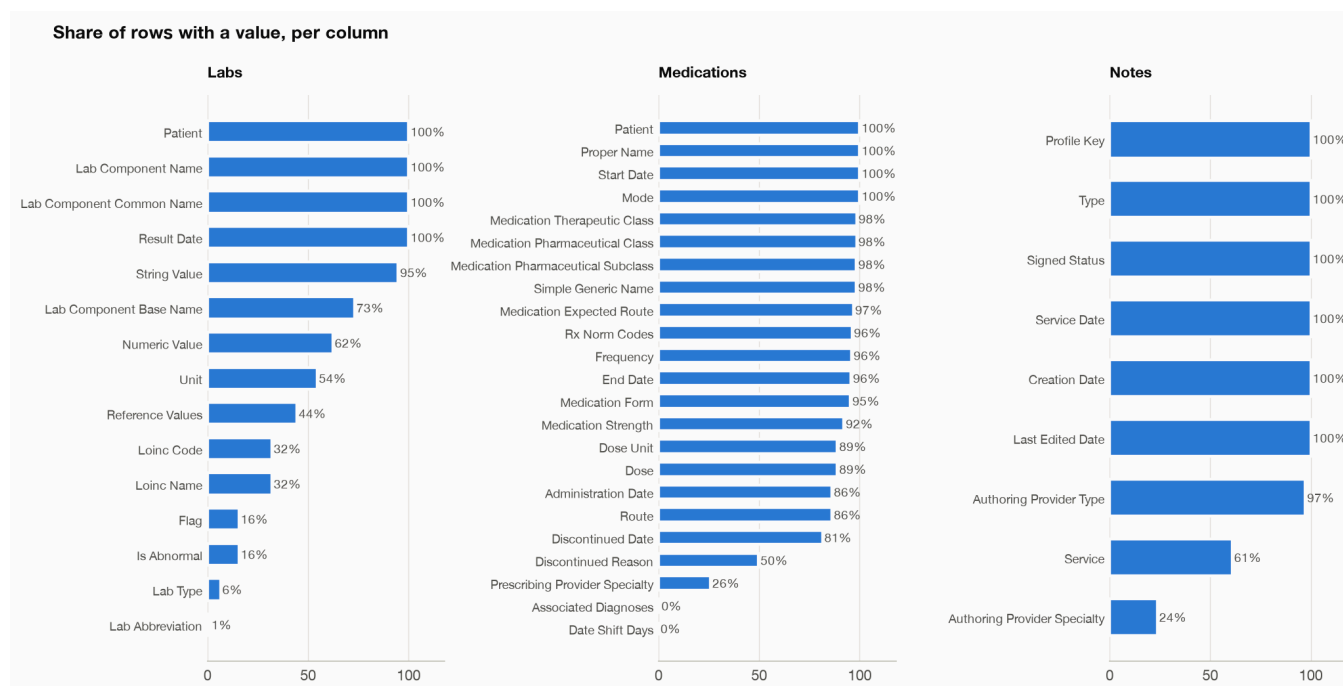
Durability is measured over years, so this matters more for the study than any other gap. For each patient it counts the distinct years with any data, and the gap between the first and last year.



63% of patients have notes from a single year. Among the 17 patients with labs and medications, the median is two years of data, but a quarter of them have eight or more.

06 How complete each column is

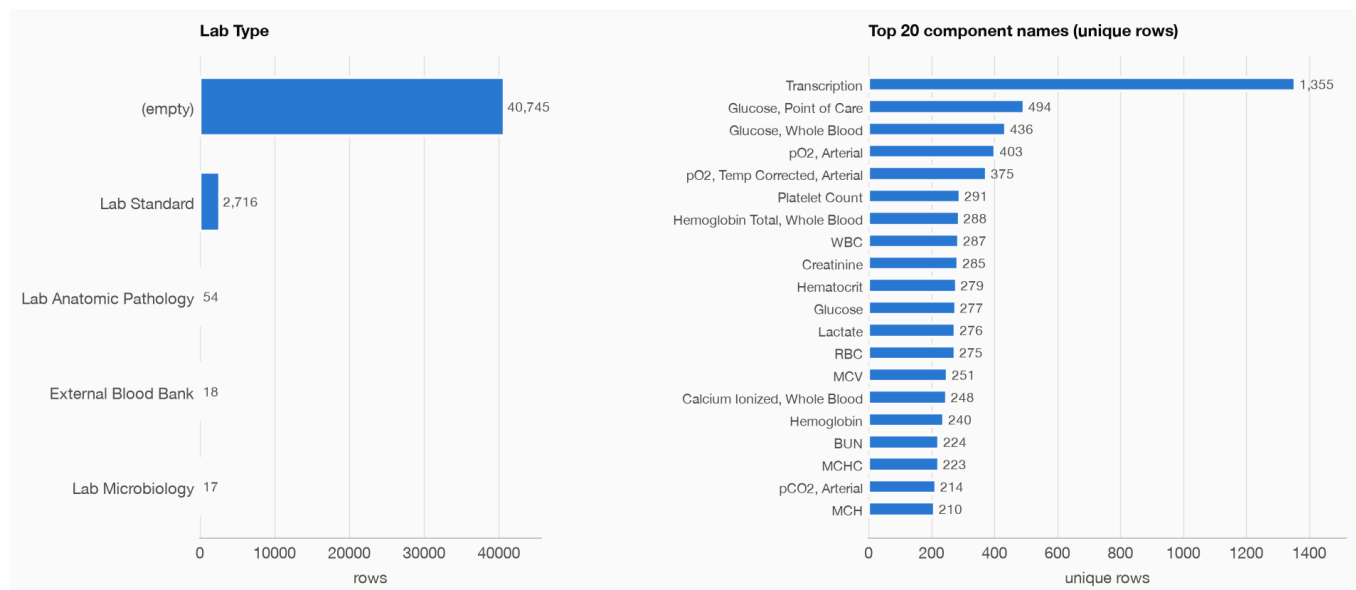
The share of rows that actually hold a value, column by column. Identifiers and dates are always filled. Many descriptive columns are not.



- **Lab results:** the lab category is filled on 6% of rows and the abnormal flag on 16%. A missing flag does not mean the result was normal; the flag is only written when a result is abnormal. Standard lab codes (LOINC) are present on 32% of rows.
- **Medications:** diagnoses linked to an order are almost never filled, and the date-shift column is empty on every row.
- **Clinical notes:** the hospital service is missing on 39% of notes and the author's specialty on 76%.

07 Lab results

The lab file holds more than lab results. Its largest single item is "Transcription", which covers free-text reports, and it also carries blood bank, pathology and device records.



Left: the lab category is blank on 40,745 rows. Right: the 20 most common test names, counted after removing repeated rows.

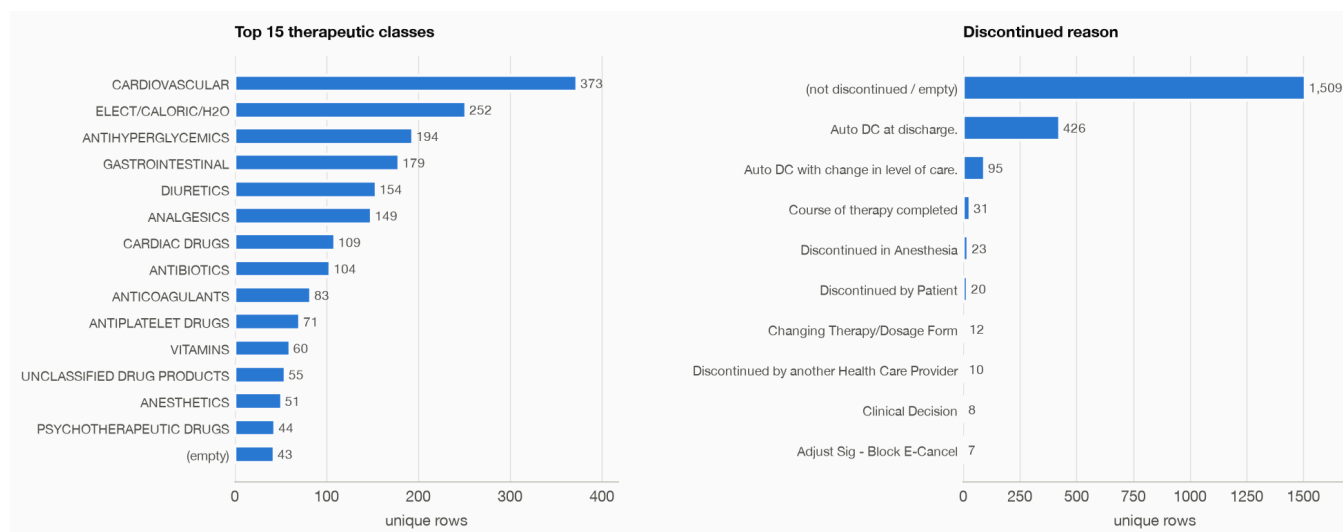
The same test appears under several names, lab codes and unit spellings. Before we can follow kidney function or calcium over time, each of these groups has to be merged into one measurement with one unit.

Test	Names used	Lab codes	Unit spellings	Patients	Distinct results
Creatinine	6	3	3	17	310
eGFR	5	2	4	17	192
Haemoglobin	5	3	2	17	582
Calcium	4	3	2	17	476
Phosphorus	2	3	2	17	213
LV ejection fraction	1	0	1	17	77

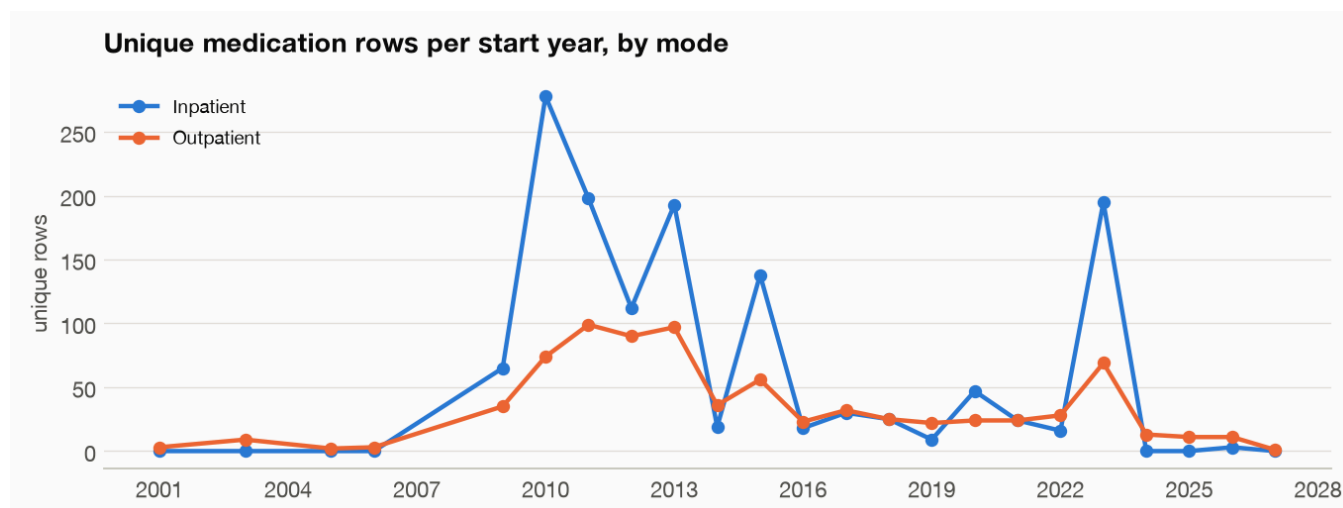
Across the whole file there are 703 test names and 76 unit spellings. 62% of rows carry a number. 273 results are written as limits such as ">60" rather than exact values.

08 Medications

Once repeated rows are removed, 2,157 medication records remain, covering 313 different drugs. 64% of them are inpatient orders.



Left: cardiovascular drugs and fluids or electrolytes lead. Anticoagulants (83) and antiplatelet drugs (71) matter most for valve durability. Right: the most common stop reason is automatic discontinuation at discharge.

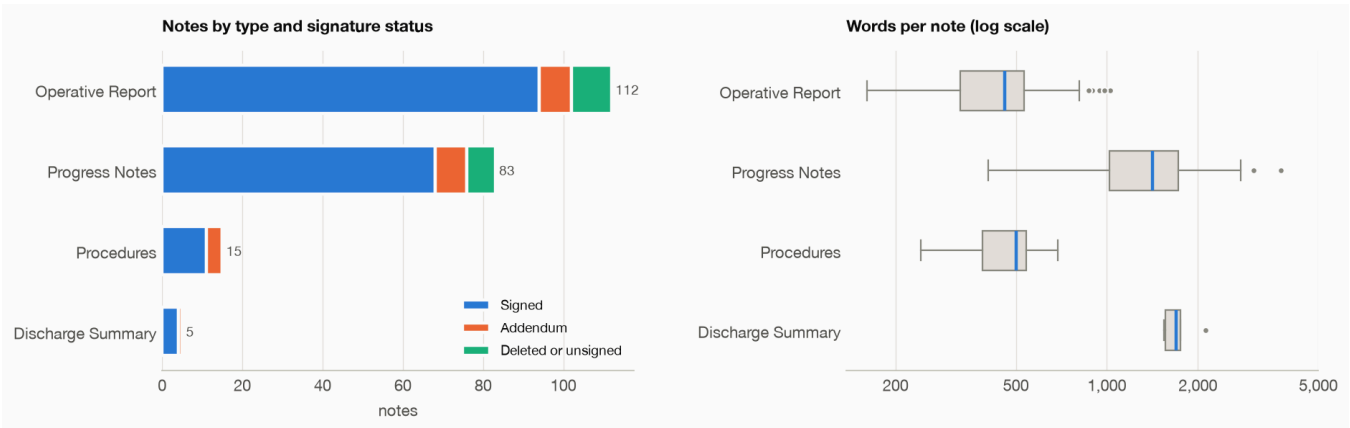


Inpatient orders spike in admission years (2010, 2013, 2015, 2023). Outpatient prescriptions, which describe long-term therapy, are a thinner and steadier line.

Because dates are years only, a start and stop in the same year tells us little about how long a patient actually took a drug.

09 Clinical notes

215 notes in four types. Operative reports are the only place that names the implanted valve and its size, and follow-up notes are the only place with echo measurements. There is no separate echo table.



Left: 13 notes are deleted drafts and 4 are unsigned, and 9 of the deleted drafts are operative reports. Right: progress notes and discharge summaries are about three times longer than operative reports. The median note has 577 words.

Names, dates and ages were replaced with placeholders. Every note has date and name placeholders, and most also hide the patient's age, so age at surgery cannot be read from the text.

Note type	Notes	Hidden dates	Hidden names	Hidden ages	Notes hiding age
Operative report	112	383	992	148	90%
Progress note	83	1,853	1,036	210	93%
Procedure note	15	32	132	11	73%
Discharge summary	5	110	133	13	100%

10 What this means for the study

The extract can show how a durability study would run from start to finish. It cannot tell us which valves last longer. Five findings drive that conclusion.

- **Two groups that never meet:** the 100 patients with an operative report have no lab or medication records, and the 17 with those records have no operative report. No patient links valve details to kidney function or drug history.
- **Echo values exist only as text:** mean gradient, valve area, DVI and regurgitation grade have to be read out of the notes. That is why we built the note extraction step before any modelling.
- **Follow-up is short:** most surgical patients have notes from one year only. Deterioration is defined as a change from the first echo after surgery, so it needs at least two echoes years apart, and few patients have them.
- **The structured files need cleaning:** more than half the rows are repeats, one test goes by up to six names, and most medication rows are orders from the hospital stay around surgery.
- **Time is measured in years:** dates are shifted and cut to the year, so we only know which year an event happened in, not the date.

Next steps: the cleaning plan (merge repeats, unify test names and units, one implant record per patient, one echo timeline per patient) is in the preprocessing notebook. What a full-size study would need, above all a death and last-contact record for every patient, goes into the data request in the data plan.